Elizabeth Micaela Herrejon

elizabeth.m.herrejon@gmail.com elizabeth-h-659b74191 eherrejon3
https://eherrejon3.github.io/grad-website.github.io/

Employment

JUL 2025 – · · · ·

Research Engineer I, Georgia Tech Research Institute - EOSL.
Supported the development and execution of state-of-the-art algorithms involving data fusion, sensor fusion, machine learning, manned/unmanned avionics improvement, and soldier protection software development projects for Department of Defense programs. Focused on different GUIs for user interactions: a robust imGui platform, a full-stack of NextJS/Django/psychopg for a custom data analysis app. Supported development of two different path planning algorithms using ROS2 and c++ wrappers to communicate with simulated plane agents.

May 2024 – JUN 2025

Peace Corps Volunteer, NR1. Janajyoti Secondary School, Chapakot Municipality, Syangja District, Nepal.
Served as a co-teacher/trainer with the Computer Science teachers at the school to enhance student learning of ICT and collaborated with CS teachers to increase their capacity in resolving ICT hardware issues. Coordinated with IT development team about efficiency controls and reviewed the school's ICT roadmap to provide technical advice for revision and improvement. Introduced design enhancements, updates, and programming changes for subsystems of systems software, including operating systems, networking, utilities, databases, applications and Internet-related tools. Restored two damaged computers and turned them into Windows Servers. Setup a secure closed loop network exclusive to the Janajyoti Administrators and changed all client PC's DNS to Server IP addresses. Added internet to all 30 computers and installed an endpoint security software to monitor, restrict, and protect internent access for student and teacher accounts.

SEP 2022 – MAY 2024

Research Engineer I, Georgia Tech Research Institute - EOSL.
Utilized React Javascript to build new GRIP website from scratch, with a goal to elevate and swiftly improve user experience. Under the Sensor Division, it was possible to work on python scripts to visualize data from HTML, CSV, etc, then plot those results to convey the data; it included algorithms for areas such as target tracking, alternative forms of positioning – navigation – timing (aPNT), and processing of sensor data as needed. Specific tasking included new feature development, debugging code, test case development, integration with other applications and data sources, and in-depth analysis.

JAN 2020 – FEB 2022

Electrical Engineering Researcher Assistant, Georgia Tech Research Institute - ATAS.
Worked with the System Design Division to support various engineers in SSD to troubleshoot circuits, create circuit diagrams, build cable assemblies, and generate technical diagrams. Received Interim Security Clearance and worked with electrical engineers to troubleshoot different chassis wiring diagrams and perform maintenance on hardware units. Used multimeters and oscilloscopes to run verification tests.

MAY 2021 – JUL 2021

Undergraduate Research Internship Program, Intern, Georgia Tech Research Institute - Robotic Behavior Development.
Worked with the creation and composition of robotic primitive skills to create flexible behaviors on a 6 degree-of-freedom robotic system in order to create dexterous grasping and manipulation behaviors with a robotic arm with integrated force/torque and vision sensing. Coded in python to develop a deep learning program, Mask RCNN, to catalog and identify different types of electrical connectors using a custom dataset with noise to run training and inference on unmarked images and videos.

Projects

These are some projects that highlight the most recent work captured from 2025 – 2026.

ICT Improvement of Rural Computer Lab

Computer Lab Manual for Nepal School

https://eherrejon3.github.io/grad-website.github.io/assets/pdfs/Computer_Lab_Admin_Master_Manual.pdf

Stun Gun Glove: *nemi*

Website for Senior Capstone Project

https://eherrejon3.github.io/SweetDreams.github.io

Dr. Eloisa Klementich Site

NextJS/Vercel Stack

https://www.dreloisaklementich.org/

Hippo Mexico Site

NextJS/Vercel Stack

https://www.hippomexico.com/

Research Presentations & Publications

Conference Proceedings

1. W. Freidank, E. Herrejon, C. Lindbeck, A. Popa, N. Boyd, and B. S., "Robotic automation of electrical maintenance," in *Robotic Development and Behavior Summer Proposal: GRIP Presentation*, 5th Floor Board Room, GTRI Headquarters, Atlanta, Georgia, July 2021, (poster and powerpoint).
2. E. Herrejon, "Gyrobot: An omnidirectional and versatile spherical robot," in *HIVES IRAD Extravaganza: GYROBot Presentation*, Atlanta, Georgia, July 2024, (poster).
3. E. Herrejon et al., "Nemi: The stun gun glove," in *Georgia Tech Product Exposition: Sweet Dreams Pitch*, Atlanta, Georgia, April 2022, (poster and powerpoint).

Manuals

1. E. Herrejon, Ed., *Janajyoti Secondary School: Computer Lab Manual*. Kathmandu, Nepal: Peace Corps Nepal, 2025.

Skills

Languages	English, Spanish, Nepali, and French.
Coding	Java, Jupyter Notebooks, Python, C/C#/C++, LaTeX, Git, Shell
Protocols	GDAL, FPGA, MQTT, FTP, Tailscale, ROS2
Libraries	Numpy, Pytorch, Pandas, Matplotlib, Dart, SLAM
Web Dev	HTML, CSS, TAILWINDCSS, JavaScript, ReactJS, NextJS, NodeJS.
Android Dev	Koitlin, Flutter, Android Studio
Softwares	National Instruments, After Effects, Sony Vegas Pro, Autocad, Inventor, Arduino, Solidworks, Multisim, Matlab, KiCad, LTSpice, Windows Server 2022/2019
Lab Equipment	DC power supplies, function generators, oscilloscopes, digital multimeters, RCL meters, etc.
Microsoft Office	Proficient in Excel, Word, Powerpoint, Visio, Publisher, Visual Studio

Education

2020 – 2022	**B.T Electrical Engineering, Minor in Robotics** Georgia Institute of Technology, Atlanta, Georgia.
2018 – 2019	**BSc Mathematics** Georgia State University, Atlanta, Georgia.

Miscellaneous Experience

Awards and Achievements

JUN 2023	**GTRI HIVES Early Career IRAD Recipient**, Funding for GYROBot.
APR 2022	**OMED Bronze Senior Excellency Award**, Georgia Tech.

Certification

NOV 2023	**UNESCO & the SDG Academy ICT Course Certificate**.
Iss. 02 15 2020	**Secret Clearance**.

References

Available on Request